Rolling challenges

Objective

These are additional, optional fun challenges to either work through during the course or after the course has finished. Learners can choose the level of difficulty from Beginner to Advanced (but not too advanced). Solutions are provided.

Beginner challenges

Grade reporting tool

Allow users to put a score of between 0 and 100 into a program. The system should then return their grade based on the following scale:

- >90 = A*
- >80 = A
- >70 = B
- >60 = C
- >50 = D
- >40 = E

Anything under this score should be reported as an F.

It should loop and allow users to put another grade in or quit the system. There should also be validation to ensure they cannot put a score above 100 or below 0 into the program.

Areas and volumes

Write a program to work out the areas of a rectangle.
Collect the width and height of the rectangle from the keyboard.

Calculate the area and display the result.

Extend this program to ask if they want to include a 3rd dimension and return the volume of the rectangular cuboid.

Speed reporting

Write a program that will work out the distance traveled if the user enters in the speed and the time.

Get the program to tell you the speed you would have to travel at in order to travel a certain distance within a certain time (as entered by the user).

Intermediate challenges

Rock, paper, scissors

Create a game of Rock, Paper, Scissors where the user plays against the computer.

Don't forget to randomize the selection from the computer.

Extension

Make sure the user inputs a valid choice.
Add a loop structure to play several times and keep score.

Password strength analyzer

Create a program which accepts a user password and tells them if it needs improving.

A strong password contains at least 1 of each of the following:

- Capital letters
- Lowercase letters
- Number
- Symbols

And must be at least 8 characters long.

The program must report what a password is missing. For example, the supplied password **LetmeInNow** should return:

- The password has no Symbols.
- The password has no Numbers.

Fibonacci sequence generator

Create a Fibonacci sequence generator.

Fun fact: The Fibonacci sequence was originally used as a basic model for rabbit population growth.

The Fibonacci sequence: 0, 1, 1, 2, 3, 5, 8, 13

The Nth term is the sum of the previous two terms. So, in the example above the next term would be 21 because it would be the previous two terms added together (8+13).

You will need to create a list of Fibonnaci numbers up to the 50th term.

The program will then ask the user for which position in the sequence they want to know the Fibonacci value for (up to 50).

For example:

Program: 'Which position in sequence?'

User: '6 (start counting at 0).'

Program: 'Fibonacci number is 8.'

Advanced challenges

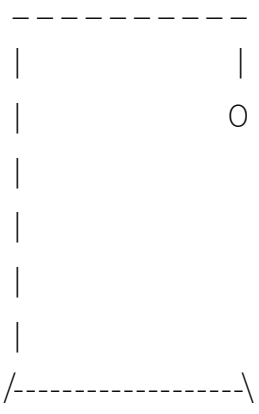
Hangman

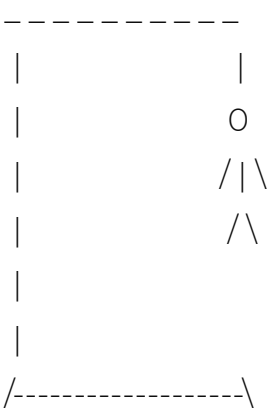
Create a game of hangman.
The program should randomly choose a word and present the user with the underscores as placeholders, for example:

_ _ _ _ _ _

It should tell them the choices they have already made and prevent the user from choosing the same letter again.

It should also display a graphic of the state of hanging, for example:





Caesar cypher

Write a program to perform a basic 'Caesar' encryption and decryption on text.

This algorithm works by moving letters along by an 'offset'.
If the offset is 2, then b —> d, h—>j, etc.

Try to write two functions — one called 'encrypt' and one called 'decrypt'. Both will return a string.

The user selects whether they wish to encrypt or decrypt.

The user enters a sentence to encrypt and the encryption key (i.e., how many we move the letters along — this is a smallish integer).

The program responds with the encrypted or decrypted version.

Rolling challenge solutions

Beginner challenges

Grade reporting tool

```
again = "yes"
while again == "yes":
    score = int(input("Enter a score between 0 and 100: "))
    if score > 100 or score < 0:
        print("That Score was out of bounds")
    else:
        if score > 90:
            print('A*')
        elif score > 80:
            print('A')
        elif score > 70:
            print('B')
        elif score > 60:
            print('C')
        elif score > 50:
            print('D')
        elif score > 40:
            print('E')
        else:
            print('F')

    more = input("Would you like to check another score")
    if more not in ("Y", "yes", "y"):
        again = "no"
print("Thank you - Goodbye")
```

Areas and volumes

```
def area(h, l):
    print(f"The area of that rectangle is {h*l} cm2")
    vol = input("Would you like to calculate the volume?
('Y')")
    if vol in ("Y", "y"):
        depth = int(input("Please enter the depth: "))
        volume(h,l,depth)
    return None

def volume(h,l,d):
    print(f"The volume of that shape is {h*l*d} cm3")
    return None

height = int(input("Please enter the height: "))
length = int(input("Please enter the length: "))

area(height,length)
```

Speed reporting

```
def calc_distance():
    speed = float(input("Enter speed in Mph per hour: "))
    time = float(input("Enter time in hours: "))
    return f"You would travel {speed * time} Mph"

def calc_required_speed():
    distance = float(input("Enter distance in Miles: "))
    time = float(input("Enter time in hours: "))
    return f"You would need to average {distance/time} Mph"


print(calc_distance())
print(calc_required_speed())
```

Intermediate challenges

Rock, paper, scissors

```
from random import randint

#create a list of play options
t = ["Rock", "Paper", "Scissors"]

#assign a random play to the computer
computer = t[randint(0,2)]

#set player to False
player = False

while player == False:
#set player to True
    player = input("Rock, Paper, Scissors?")
    if player == computer:
        print("Tie!")
    elif player == "Rock":
        if computer == "Paper":
            print("You lose!", computer, "covers", player)
        else:
            print("You win!", player, "smashes", computer)
    elif player == "Paper":
        if computer == "Scissors":
            print("You lose!", computer, "cut", player)
        else:
            print("You win!", player, "covers", computer)
    elif player == "Scissors":
        if computer == "Rock":
            print("You lose...", computer, "smashes", player)
        else:
            print("You win!", player, "cut", computer)
    else:
        print("That's not a valid play. Check your spelling!")
    #player was set to True, but we want it to be False so the
loop continues
    player = False
    computer = t[randint(0,2)]
```

Password strength analyzer

```
nums = 0
uppers = 0
lowers = 0
symbols = "!£$%^&*()@#~?><,./{;:'[]}"
syms = 0

pw = input("Please enter your password: ")
if len(pw) < 8:
    print("Your password is too short.")
else:
    for char in pw:
        if char.isnumeric():
            nums += 1
        elif char.isupper():
            uppers += 1
        elif char.islower():
            lowers += 1
        elif char in symbols:
            syms += 1
        else:
            print("Your password contains invalid characters.")
            break
if nums == 0:
    print("Your password does not contain any numbers.")
if uppers == 0:
    print("Your password does not contain any uppercase
letters.")
if lowers == 0:
    print("Your password does not contain any lowercase
letters.")
if syms == 0:
    print("Your password does not contain any symbols.")
```

Fibonacci sequence generator

```
n = int(input("Which number of the sequence do you want? ")) -2
num1 = 0
num2 = 1
next_number = num2
count = 1

while count <= n:
    count += 1
    num1, num2 = num2, next_number
    next_number = num1 + num2

print(next_number, end=" ")
```

Advanced challenges

Hangman

```
import random

# List of words for the game
words = ["hangman", "python", "game", "programming", "openai"]

# Hangman graphics
hangman_graphics = [
    '''
     +---+
         |
         |
         |
        ===
    ''',
    '''
     +---+
     O   |
         |
         |
        ===
    ''',
    '''
     +---+
     O   |
     |   |
         |
        ===
    ''',
    '''
     +---+
     O   |
    /|   |
         |
        ===
    ''',
    '''
     +---+
     O   |
    /|\  |
         |
        ===
    ''',
    '''
     +---+
     O   |
    /|\  |
    /    |
        ===
    ''',
    '''
     +---+
     O   |
    /|\  |
    / \  |
        ===
    '''
]
```

```
def select_word():
    """Selects a random word from the word list."""
    return random.choice(words)

def initialize_progress(word):
    """Initializes the progress string with underscores."""
    return "_" * len(word)

def update_progress(word, progress, letter):
    """Updates the progress string with correctly guessed letters."""
    updated_progress = ""
    for i in range(len(word)):
        if word[i] == letter:
            updated_progress += letter
        else:
            updated_progress += progress[i]
    return updated_progress

def display_progress(progress, guesses_left):
    """Displays the hangman graphics and the current progress."""
    print(hangman_graphics[6 - guesses_left])
    print("Progress:", progress)

def hangman():
    word = select_word()
    progress = initialize_progress(word)
    guesses_left = 6
    guessed_letters = []

    print("Welcome to Hangman!")
    display_progress(progress, guesses_left)

    while guesses_left > 0:
        guess = input("Guess a letter: ").lower()

        if len(guess) != 1 or not guess.isalpha():
            print("Invalid guess. Please enter a single letter.")
            continue

        if guess in guessed_letters:
            print("You have already guessed that letter. Try again.")
            continue

        guessed_letters.append(guess)

        if guess in word:
            progress = update_progress(word, progress, guess)
            display_progress(progress, guesses_left)

            if progress == word:
                print("Congratulations! You guessed the word correctly!")
                break
        else:
            guesses_left -= 1
            display_progress(progress, guesses_left)
            print("Wrong guess. You have", guesses_left, "guesses left.")

    if guesses_left == 0:
        print("You lost! The word was:", word)


hangman()
```